Auditable Decision-Making in Intelligent Cyber-Physical Systems Using Semantic Transition and Rough Sets

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Presentation Map

1. Motivation
2. Method
3. Experimental Results
4. Discussion

Why Auditing Is Needed

CPS safety gap

- Deep controllers are accurate but opaque.
- Local XAI artifacts do not provide reusable global decision logic.
- Safety-critical systems need verifiable reasoning and failsafe behavior.

Study goal

Translate latent neural features into explicit, conflict-aware symbolic rulebooks without retraining the underlying controller.

Keywords: Explainable AI, cyber-physical systems, rough sets, model auditing, production rules, semantic transition.

Core Contributions

1. Transition

Linear transition matrix maps compressed deep representations into physically interpretable semantic attributes.

2. Discretization

WEDD balances entropy reduction with local density to convert continuous signals into robust symbolic states.

3. Rulebook

Rough-set induction creates a global rulebook with explicit conflicts, boundary regions, and abstentions.

The framework acts as a non-destructive post-hoc audit layer for existing neural controllers.

Position Relative to Prior Work

What existing methods provide

- Local saliency and attribution support model inspection.
- Concept bottlenecks align activations with human variables.
- Decision trees and rule learners provide symbolic branches.

What remains missing

- Discrete symbolic layer for global logical reasoning.
- Formal abstention when evidence is contradictory.
- Auditable bridge from neural features to CPS-relevant semantics.

SEMTRA Pipeline

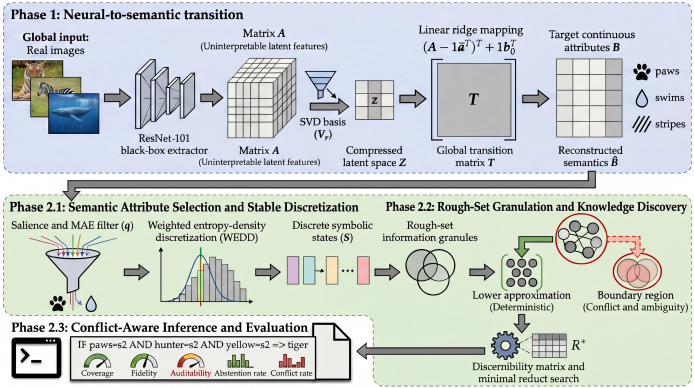


Figure 1: Conceptual SEMTRA architecture: from the latent space of a neural network to a verified rulebook.

Two-stage audit layer

- Frozen neural feature extractor.
- Neural-to-semantic transformation.
- Semantic-to-symbol granulation.
- Rulebook plus audit trace.

Phase 1: Neural-to-Semantic Transition

Ridge transition operator

$$A \in \mathbb{R}^{m \times k}, \qquad B \in \mathbb{R}^{m \times l}$$

$$T = (A^\top A + \alpha I)^{-1} A^\top B$$

Why it is auditable

- Each weight links latent evidence to a semantic attribute.
- The base controller remains frozen.
- Engineers can inspect which features drive physical meanings.

The manuscript reports this as the bridge between latent space and physically meaningful semantic variables.

Phase 2.1: Stable Discretization (WEDD)

Equation (1): threshold objective

$$J_j(\theta) = \lambda_H \widetilde{H}_j(\theta) + (1 - \lambda_H)\widetilde{p}_j(\theta) \qquad (1)$$

$\widetilde{H}_j(\theta)$ is information entropy; $\widetilde{p}_j(\theta)$ is a local density proxy estimated with KDE.

Design effect

- Thresholds move toward sparse semantic regions.
- Micro-perturbations are less likely to change symbolic state.
- The resulting states are suitable for rough-set granulation.

Phase 2.2: Knowledge Granulation

Equation (2): induced production rule

$$r : \bigwedge_{j \in R_r} [s_j = v_j] \Rightarrow d = y_r. \qquad (2)$$

Objects with identical semantic signatures but different classes form boundary regions.

Compact audit artifact

- Indiscernibility relation over discretized states.
- Lower approximation induces reliable rules.
- Minimal reducts preserve logic with fewer attributes.

Conflict-Aware Inference and Auditing

Honest silence

- Conflict: activated rules disagree.
- Abstention: no reliable rule applies.
- CPS response: failsafe or human review.

Aggregate support score

$$S(y_k) = \sum_{r \in \mathcal{A}(x),\, y_r = y_k} \mathrm{conf}(r)\,\mathrm{supp}(r)$$

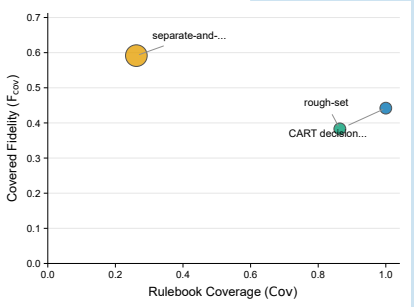


Figure 2: Trade-off between rulebook coverage and fidelity. SEMTRA occupies a high-coverage region while explicitly reporting conflicts.

Semantic-Space Reconstruction

AwA2 Protocol A

The evaluation uses Animals with Attributes 2: 50 object classes described by 85 semantic attributes, averaged across five seeds.

Table 1: AwA2 Protocol A transition reconstruction metrics. Metrics are reported as mean $\pm$ standard deviation across five seeds.

Split	MAE	RMSE	Mean semantic correlation
train	0.1027 ± 0.0003	0.1482 ± 0.0004	0.8079 ± 0.0009
validation	0.1030 ± 0.0003	0.1486 ± 0.0005	0.8065 ± 0.0014
test	0.1029 ± 0.0005	0.1484 ± 0.0007	0.8070 ± 0.0018

Test MAE $= 0.1029$ and correlation $= 0.8070$ show a stable neural-to-semantic bridge.

Rulebook Properties and Audit Tax

Table 2: General rulebook properties and symbolic-baseline comparison. Point estimates are reported alongside standard deviations over five random seeds for the proposed method.

Metric	Value (ours)	CART baseline	Separate-and-conquer
Number of rules	59.00 ± 2.55	53	197
Test coverage	0.8480 ± 0.0530	1.0000	0.2620
Covered accuracy	0.3973 ± 0.0109	0.4129	0.6334
Covered fidelity to base	0.4048 ± 0.0078	–	–
Average conditions per rule	4.0397 ± 0.0543	7.0000	3.2589
Test abstention rate	0.1520 ± 0.0530	0.0000	0.7380
Test conflict rate	0.1375 ± 0.0532	0.0000	0.2301

The audit tax is the part of model behavior that can be preserved as strict, inspectable predicates.

Robustness to Semantic Noise

Table 3: Synthetic stress-test results averaged over random seeds.

Noise (σ)	MAE	Threshold err.	Rule Jaccard	Macro-F1	Coverage
0.000	0.000	0.014	0.700	0.879	0.971
0.100	0.021	0.018	0.763	0.881	0.962
0.200	0.041	0.017	0.743	0.838	0.952

Result

- Macro-F1 remains 0.838 at $\sigma = 0.20$.
- Coverage remains 95.2%.
- WEDD filters small semantic perturbations while preserving inference.

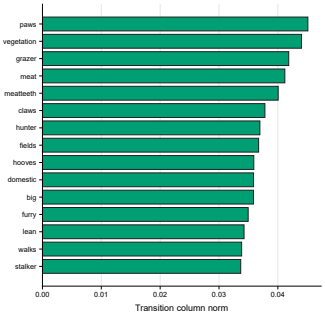
Representative Production Rules

Table 4: Representative induced production rules from AwA2 Protocol A. Support refers to the number of instances covered in the training partition.

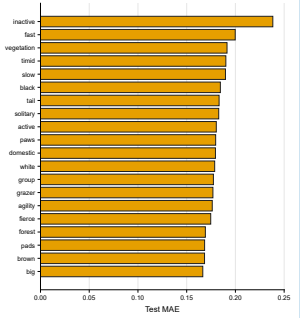
Rule / class	Antecedent	Numeric bounds	Support / Conf.
R0001 / tiger	paws=s2 AND hunter=s2 AND small=s0 AND yellow=s2	paws≥ 0.40; hunter≥ 0.40; small< 0.15; yellow≥ 0.40	115 / 0.896
R0003 / antelope	stripes=s1 AND hooves=s3 AND paws=s0 AND hands=s0 AND small=s1 AND yellow=s1	stripes$\in [0.15, 0.40]$; hooves≥ 0.70; paws< 0.15; hands< 0.15; small$\in [0.15, 0.40]$; yellow$\in [0.15, 0.40]$	132 / 0.871
R0004 / zebra	hunter=s0 AND water=s0 AND small=s0 AND jungle=s2 AND yellow=s1	hunter< 0.15; water< 0.15; small< 0.15; jungle≥ 0.40; yellow$\in [0.15, 0.40]$	434 / 0.866

Each rule is inspectable against physical or semantic sensor specifications.

Semantic Attribute Diagnostics



(a) Transition salience by semantic attribute.



(b) Attribute-wise semantic reconstruction error on AwA2.

Figure 3: Semantic attribute diagnostics: (a) transition salience by semantic attribute; (b) attribute-wise semantic reconstruction error on AwA2.
High-salience attributes such as paws and vegetation reveal which physical features dominate model logic.

Threshold Optimization Dashboard

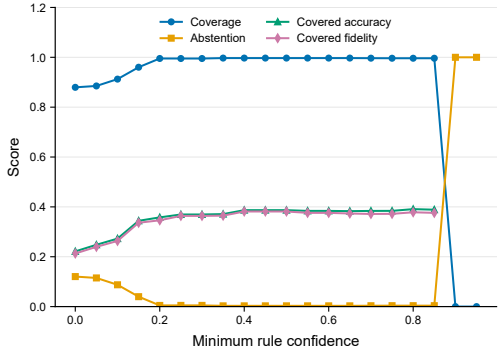


Figure 4:

Rulebook and threshold optimization. Coverage, abstention, covered accuracy, and covered fidelity as the minimum rule-confidence threshold varies.

Operational tuning

- Lower threshold: more autonomous coverage.
- Higher threshold: stricter human oversight.
- The dashboard exposes the coverage-abstention trade-off directly.

Inference Traces

Table 5: Representative rule inference traces with object ID, prediction, matched-rule evidence, and antecedent states.

Case / object	True / pred. / mode	Rule evidence	Antecedent states
correct exact / 5083	collie / collie / exact	R0027; support 1675; conf. 0.208	stripes=s0; hooves=s0; swims=s1; paws=s3; longneck=s0; hunter=s1; ocean=s1; quadrapedal=s3
abstention / 24024	ox / abstain / exact	R0029; support 2820; conf. 0.195	stripes=s0; hooves=s2; swims=s1; paws=s0; longneck=s1; hunter=s0; ocean=s1; quadrapedal=s2
fallback error or boundary / 16781	hamster / mole / fallback	no matched rule; boundary evidence	stripes=s0; hooves=s1; swims=s1; paws=s2; longneck=s0; hunter=s0; ocean=s1; quadrapedal=s2

The trace shows whether an error comes from sensor evidence, boundary matching, or conflict in the knowledge base.

Implications for Intelligent CPS

Transparent twin

SEMTRA builds a symbolic digital twin of model logic aligned with trustworthy-AI accountability needs.

Failsafe signal

Detected conflicts 13.75% and abstentions 15.20% become explicit triggers for halt or human review.

Legacy retrofit

The method runs post hoc on frozen feature extractors, avoiding expensive controller retraining.

The covered-fidelity score of 40.48% quantifies how much neural behavior is preserved as certifiable symbolic logic.

Limitations and Future Work

Current limitations

- Static, manually annotated semantic dictionaries.
- Parallel diagnostic tool, not a full neural-network replacement.
- Interpretability requires accepting the audit tax.

Next steps

- Automated concept discovery.
- Dynamic construction of attribute spaces.
- LLM-assisted semantic representation generation for changing industrial environments.

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